Experiments and Field Experiments

Empirical Methods — Week 2

Teaching deck draft

Week 2 question

- What does randomization identify?
- What remains hard after random assignment?
- How do compliance, spillovers, implementation, scale, and equilibrium change interpretation?

Experiments are a design for building a counterfactual, not a ritual that ends the research argument.

What randomization identifies

$$Z_i \perp \{Y_i(1), Y_i(0), X_i\}$$

$$\widehat{\tau}_{ATE} = \frac{1}{N_1} \sum_{i:Z_i=1} Y_i - \frac{1}{N_0} \sum_{i:Z_i=0} Y_i$$

- Assignment makes treated and control groups comparable in expectation.
- With full compliance and no interference, the mean contrast identifies the experimental ATE.
- Covariates can improve precision; randomization is the source of identification.

Assignment is not receipt

- Z_i: randomized offer, encouragement, eligibility, information, or price.
- D_i: realized take-up or treatment receipt.
- If $D_i \neq Z_i$, the clean design object is the effect of assignment.

Field experiments often randomize what researchers or policymakers can actually control: invitations, access, timing, caseworker scripts, prices, signals, or platform rules.

ITT, take-up, and LATE

$$ITT_Y = \mathbb{E}[Y_i \mid Z_i = 1] - \mathbb{E}[Y_i \mid Z_i = 0]$$

$$ITT_D = \mathbb{E}[D_i \mid Z_i = 1] - \mathbb{E}[D_i \mid Z_i = 0]$$

$$\tau_{LATE} = \frac{ITT_Y}{ITT_D}$$

- ITT is often the policy object when the policy is an offer.
- The first stage measures how much assignment moves receipt.
- Wald/LATE is a complier effect under relevance, exclusion, and monotonicity.

Duflo–Saez as design anchor

- Randomized encouragement to attend a retirement-plan benefits fair.
- Outcomes: attendance and retirement-plan participation.
- Design lessons:
  - assignment differs from attendance;
  - peer exposure can be part of the mechanism;
  - ITT, first stage, and spillovers answer different questions.

Audit and correspondence experiments

- Randomize signals in real screening environments.
- Bertrand–Mullainathan: randomize names on resumes and measure callbacks.
- Strength: isolates employer response to a signal at initial screening.
- Limit: does not identify all hiring, bargaining, productivity, or equilibrium effects.

Cluster randomization and spillovers

$$Y_i(\mathbf{Z}) = Y_i\left(Z_i, G_i(\mathbf{Z}_{-i})\right)$$

- Cluster randomization matches treatments delivered through firms, schools, offices, villages, or markets.
- Power depends heavily on the number of clusters and within-cluster correlation.
- Spillovers can contaminate a simple contrast or become the mechanism of interest.
- Saturation designs help separate direct effects from exposure effects.

Firm and platform field experiments

Paper role	What is randomized	Design lesson
Bloom et al.	Management consulting in firms	Implementation and organizational response matter.
Pallais	Platform information / reputation	Digital markets can reveal information frictions.
Crepon et al.	Labor-market policy exposure	Direct gains can coexist with displacement.

External validity, scale, equilibrium

- Experimental population: who entered the study?
- Treatment intensity: is the pilot treatment the scalable policy?
- Institution: would delivery work outside this partner or platform?
- Market response: do prices, queues, vacancies, referrals, or competitors adjust?
- Welfare: are untreated workers or firms affected by the rollout?

Robustness is design evidence

- Baseline balance and assignment protocol.
- Compliance, take-up, and first stage.
- Attrition and outcome measurement by assignment arm.
- Correct inference for individual, cluster, or market assignment.
- Spillover and exposure diagnostics.
- Power and minimum detectable effects for meaningful outcomes.
- Ethics, implementation logs, and reproducible randomization code.

Research Lab design

Reproduce

- Synthetic Duflo–Saez-style encouragement data
- ITT, first stage, Wald/LATE
- Naive receipt contrast

Diagnose

- Balance
- Take-up
- Attrition
- Peer exposure

Transfer

- Labor-market saturation design
- Direct effects
- Spillovers
- Market-level interpretation

Takeaways

- Randomization solves assignment, not implementation or interpretation.
- ITT is the clean effect of assignment and often the policy-relevant object.
- LATE is local to compliers when encouragement moves receipt.
- Spillovers, clustered assignment, and equilibrium can change the estimand.
- Next week: DID and event studies build counterfactuals from time paths rather than random assignment.